14 Trees Foundation

BIODIVERSITY REPORT

2023-24

Ecological Society Pune

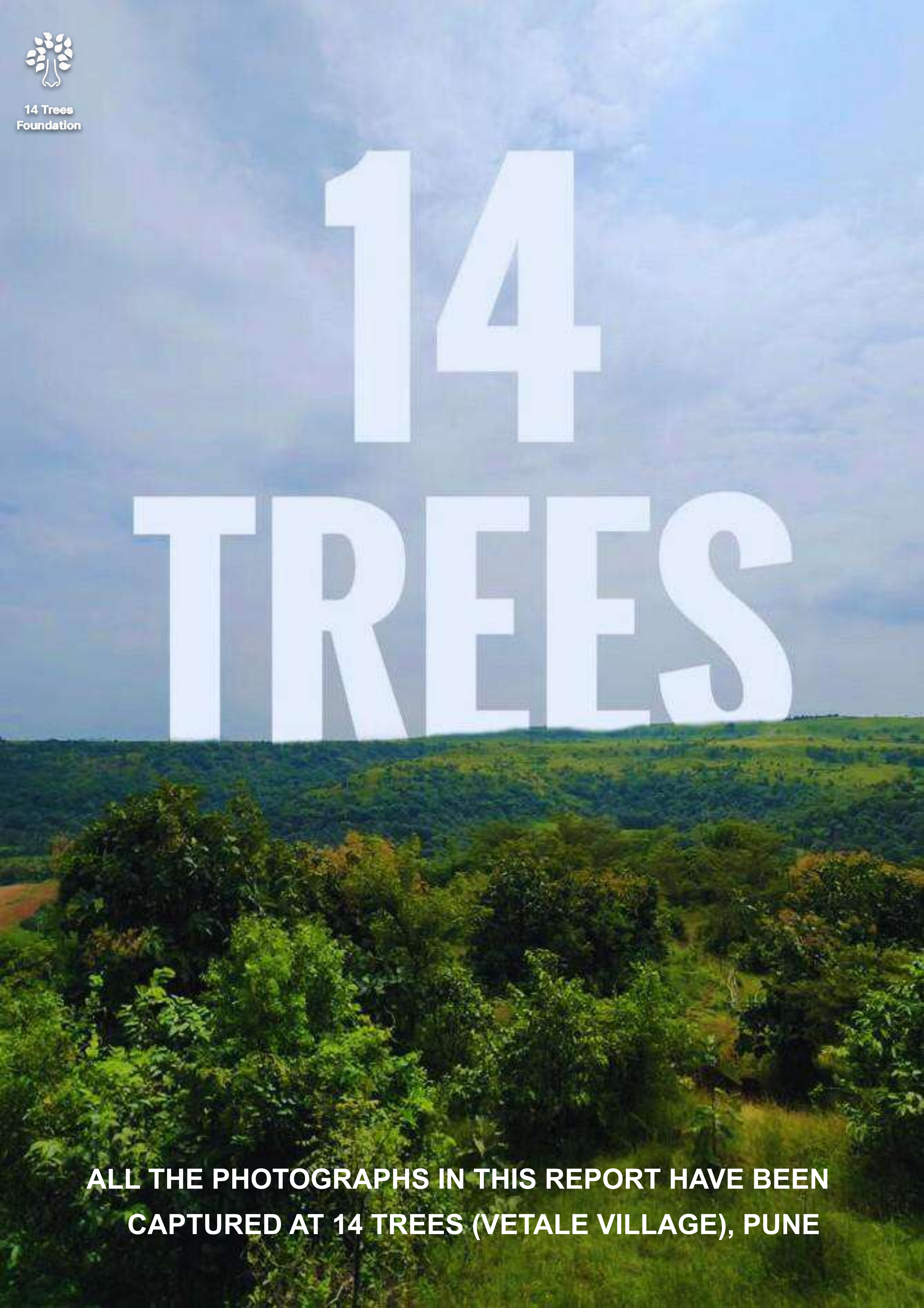
14 Trees Foundation

CONTENTS

Acknowledgement

The report has been prepared under the guidance of the 14 Trees Biodiversity Team: Mr Sanjeev Jagtap, Mr Sanjeev Naik and Ms Nirmala Samant of the CSR Team. We thank the Ecological Society's biodiversity experts Mr Pratik Purohit and Ms Revati Gindi to have carried out the survey and share the report for the year 2023-2024.

We also acknowledge guidance from Dr Kedar Bhagwat (Ecological Society), Dr Gurudas Nulkar, Ketaki Ghate and Manasi Karandikar of Oikos. This report was integrated, designed and prepared by Ms Shalvi Pawar (Biodiversity Team, 14 Trees).


14 Trees
Foundation
14
TREES
ALL THE PHOTOGRAPHS IN THIS REPORT HAVE BEEN CAPTURED AT 14 TREES (VETALE VILLAGE), PUNE

EXECUTIVE SUMMARY

This report presents the approach to biodiversity at 14 Trees, the interactions of its activities in terms of conservation and restoration. 14 Trees has published this Biodiversity report 2023-24 to provide stakeholders with clear information about the foundation's efforts to restore ecosystems and conserve biodiversity.

Biodiversity dwells in the roots of 14 Trees. For the last 10 years, the foundation has been restoring degraded barren lands in Vetale village of Maharashtra with a vision to generate native biodiverse forests for the planet's and people's well-being. Biodiversity is crucial for both humans and nature due to its numerous benefits and contributions. Biodiversity plays a crucial role in carbon sequestration, which is the process of capturing and storing atmospheric carbon dioxide (CO_2) in various carbon sinks, such as forests, soils, and oceans.

Biodiverse ecosystems are more resilient to environmental disturbances, such as climate change, pests, diseases, and extreme weather events. It also influences feedback mechanisms that regulate carbon cycling and storage. Hence, keeping this at the core of our values we are ecologically restoring the especially those degraded lands that were burnt frequently. Thus, moving beyond plantations, we are regenerating the ecological aspects and based on that, redefining the social and economical aspects.

SURVEY METHODOLOGY

In order to comprehend the biodiversity and related changes/ developments, separate survey techniques were used for flora and fauna. By employing different survey techniques for flora and fauna, the biodiversity experts aimed to gather comprehensive data on the biodiversity of the 14 Trees area, understanding of the ecological dynamics within the study area.

The survey was conducted by the biodiversity experts of Ecological Society (Pune) once every month from April 2023-March 2024. The sampling technique employed was 'belt transect' for fauna and 'random sampling' for flora. 14 Trees has diverse topography which likely contributes to the presence of diverse ecosystems like grassland, open scrub, mixed-deciduous forest, streams, ponds and valley bottoms with farm fields. Each of these ecosystems likely supports different species, contributing to the overall biodiversity of the area.

Additionally, in order to understand the year on year ecological changes and development, the 'quadrat' sampling has been adopted from this year onwards. This data will help compare the seasonal and yearly changes of selective regions/land-class. This will be helpful after two years of data is recorded in order to do comparative analysis. Hence, this will be included in the consecutive years' biodiversity reports. The method for faunal survey remains the same.

BIODIVERSITY SURVEY AND PHOTO DOCUMENTATION AT 14 TREES BY ECOLOGICAL SOCIETY (PUNE) AND VOLUNTEERS

14 Trees Foundation

TOPOGRAPHY AT 14 TREES

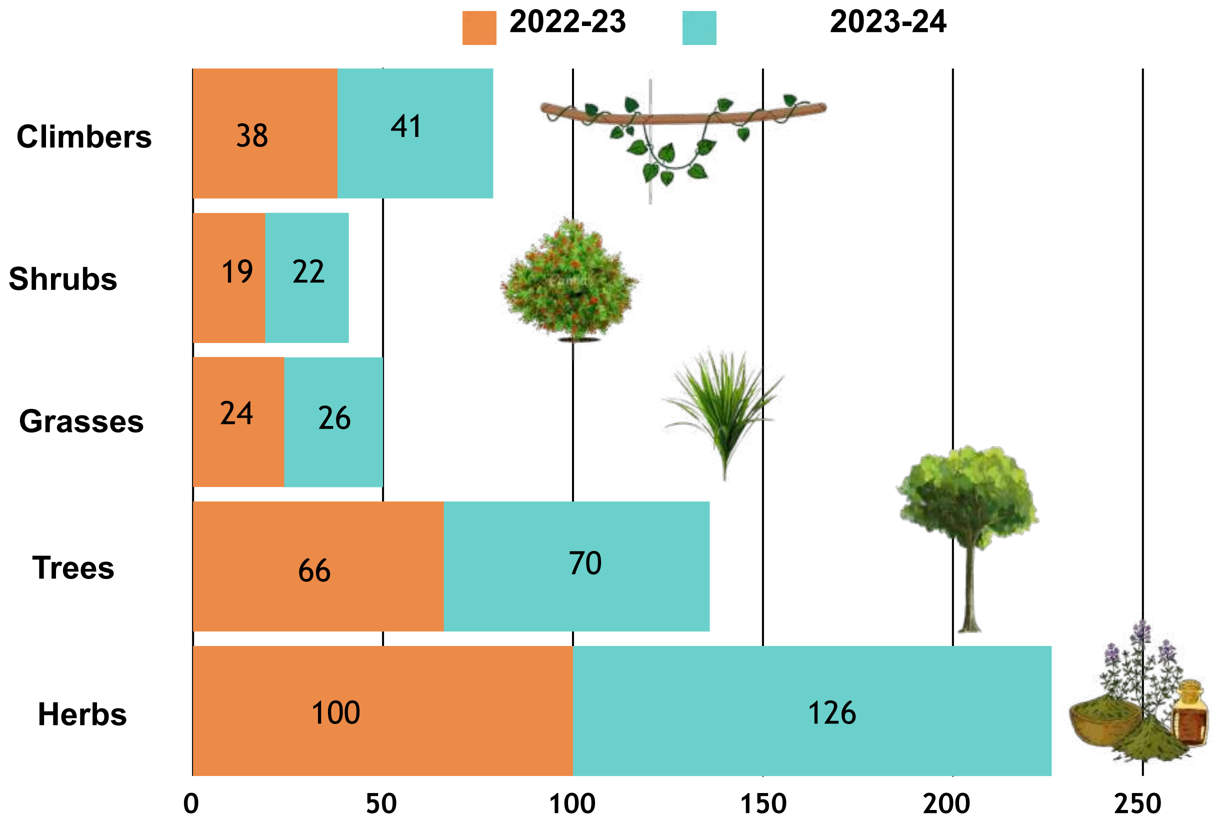
14 Trees Foundation

HABITATS AT 14 TREES

PART A
FLORA OF 14 TREES

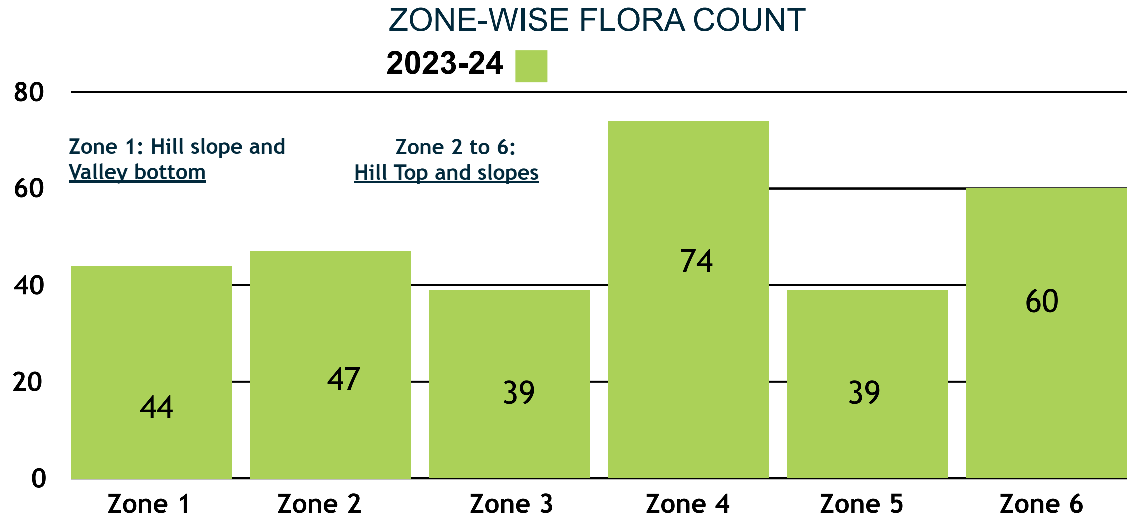
FLORA OF 14 TREES

VARIETY OF PLANTS AT 14 TREES: HABIT-WISE CLASSIFICATION YEAR 2022 -23 AND 2023-24



Total Species 2022-23: 247

Total Species 2023-24: 285



Zones 2-6 are characterised by open-scrub vegetation. Totally barren land now is dominated with maximum grass cover associated with herbs and scattered trees, climbers and shrubs. Apluda-Themeda community is replacing the original barren land grasses like Heteropogon sp.(Kali Kusal),Aristida sp.(Pandhari Kusal) indicating soil-moisture conservation and improving habitat conditions.

A) FLORAL SPECIES DISCOVERED

The new species discovered in the total 265 species observed throughout April 2023 to March 2024 are as below. All the species found are native unless mentioned.

(Format: Common local name followed by scientific name in italics)

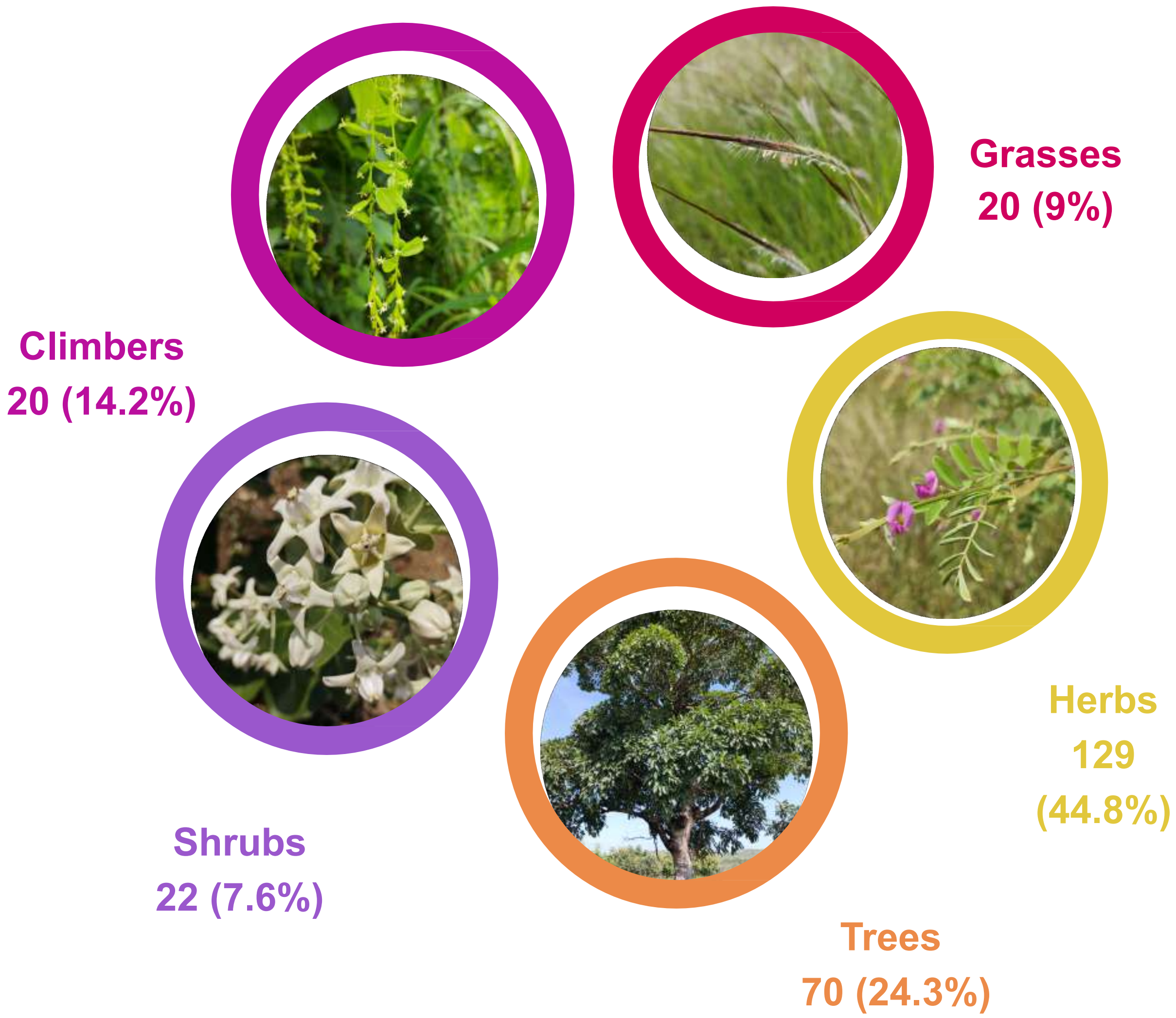
	FLORAL SPECIES NAME	HABIT		FLORAL SPECIES NAME	HABIT
1.	*Erigeron bonariensis* (introduced)	Herb	20.	Raanjire *Pimpinella tomentosa*	Herb
2.	*Solanum nigrum*	Herb	21.	Morpankhi *Strobilanthes pavala*	Herb
3.	Aghada *Achyranthus sp.*	Herb	22.	Bat davana *Conyza stricta*	Herb
4.	Sahdevi *Cyanthillium cinereum*	Herb	23.	Burandi *Blumea lacera*	Herb
5.	*Blumea mollis*	Herb	24.	Bhadali *Setaria pumila*	Herb
6.	Dudhi *Euphorbia hypericifolia*	Herb	25.	Kalmashi *Rostellularia procumbens*	Herb
7.	Ghati Pitpapra *Rungia repens*	Herb	26.	Darp tulas *Mesosphaerum suaveolens*	Herb
8.	*Tridax procumbens* (introduced)	Herb	27.	Lahan pitmari *Tylophora dalzelii*	Climber
9.	Lal Shevara *Alysicarpus vaginalis*	Herb	28.	Raan Kulith *Cajanus scarabaeoides*	Climber
10.	Vilayati takala *Senna uniflora (introduced)*	Herb	29.	Ivali bhovari *Ipomoea triloba*	Climber
11.	Hamata *Stylosathes fruticosa*	Herb	30.	Karvand *Carrisa spinarum*	Shrub
12.	Adhomukhi *Trichodesma inaequale*	Herb	31.	*Grewia abutilifolia*	Shrub
13.	*Desmodium sp.*	Herb	32.	Henkal *Maytenus senegalensis*	Shrub
14.	Popati *Physalis minima*	Herb	33.	*Pseudanthistiria heteroclita*	Grass
15.	Bala *Sida cordifolia*	Herb	34.	*Triplopogon ramosissimus*	Grass
16.	Math *Amaranthus viridis*	Herb	35.	Shindi *Phoenix sylvestris*	Tree
17.	*Justicia glauca*	Herb	36.	Sonkhair *Senegalia polyacantha*	Tree
18.	*Oxalis corniculata*	Herb	37.	Tambat *Flacourtia indica*	Tree
19.	Mhatar *Sonchus asper*	Herb	38.	Kala palas *Ougeinia oojeinensis*	Tree

Each season presents its unique challenges and opportunities for plants and fauna, influencing their survival strategies and behaviours. The new floral species was observed: **Herbs- 26, Shrubs-3, Climbers-3, Grasses-2 and Trees- 4.**

Protection and plantation activity accelerated the ground vegetation,soil moisture retention etc in the area. Addition of biomass, leaf litter making the soil more porous and fertile.Small pits and trenches on contours having grasses and herbaceous growth.Associated with this mulching activity using local grass cover helping in soil-moisture conservation.Now growth of moisture loving herbs like Akkalkara (Acmella sp.), Gangotra (Cyathocline purpurea) indicating moisture retention is taking place till December.

B) CLASSIFICATION: FLORA HABIT

TOTAL NUMBER OF SPECIES OF TYPES OF FLORA AT 14 TREES



IMPORTANCE OF FLORAL DIVERSITY

The above diagram reflects the total number of the 5 types of plant habits observed at the project site.

Diversity in flora itself is crucial for certain components of an ecosystem. Trees along with other **habits**(type pf plant) like shrubs, herbs, grasses and climbers cumulatively enrich the various properties of the landscape/ecosystem:

Shrubs and herbs contribute by alleviating soil acidification, increasing soil microbial activity and enhancing litter decomposition. The changes in soil physicochemical properties, microbial activities and biogeochemical cycles due to the presence of living roots is termed the rhizosphere effect. Additionally, **climbers** provide important habitats for a variety of animal species for nesting, shelter and foraging. The **grasses** have their own way of maintaining the ecological balance. They contribute by preserving soils, nutrients, percolating rainwater, controlling agricultural pests and habitat creation, to name a few.

ENDEMIC FLORA AT 14 TREES

Amongst the 265 floral species, some of the endemic species found at 14 Trees. **Endemic species** are the ones found only in a certain defined geographical region requiring special conditions. Hence, they are comparatively vulnerable and need conservation.

Endangered species are the ones which are at a risk of extinction due to various factors. hence, both need conservation. A few of them are mentioned below:

Purple morning glory
Argyreia cuneata
(Endemic)

Kali Musli
Curculigo orchioides
Conservation Status: Endangered

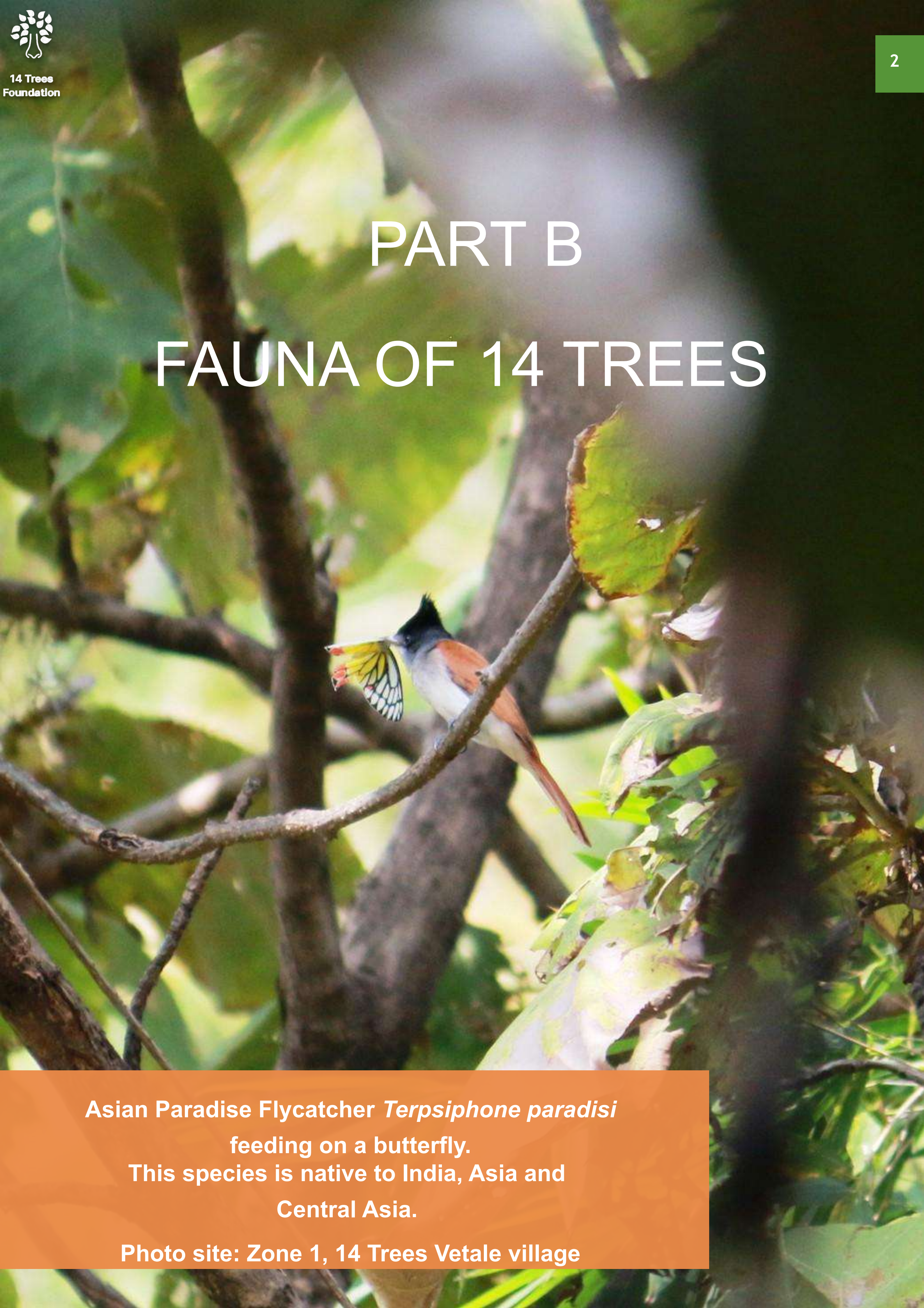
Sontikli
Pulicaria wightiana
(Endemic)

Bhui gend
Lepidagathis cristata
(Endemic)

Three-Teeth Beardgrass
Lophopogon tridentatus

PART B

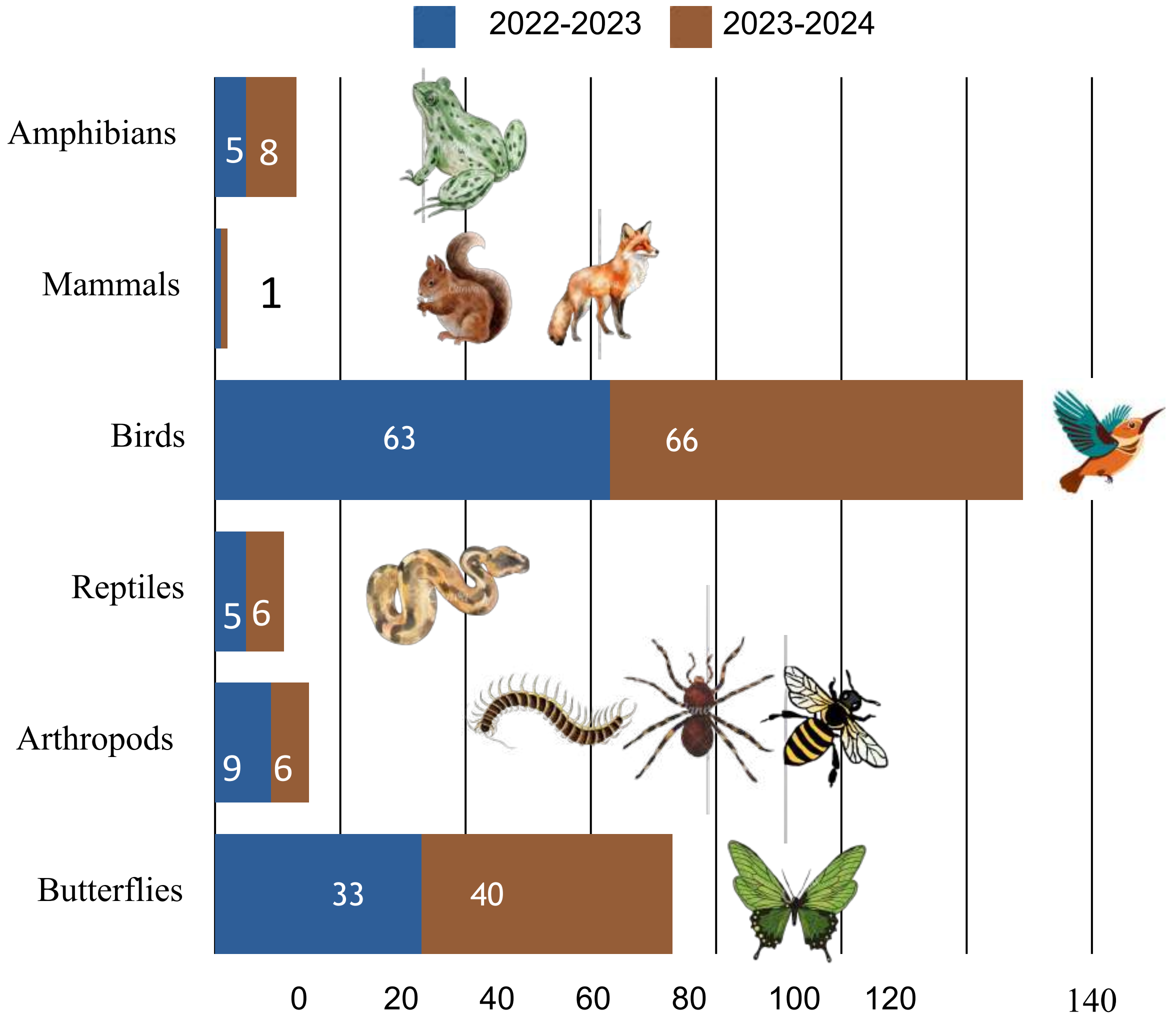
FAUNA OF 14 TREES



Asian Paradise Flycatcher *Terpsiphone paradisi* feeding on a butterfly.
This species is native to India, Asia and Central Asia.

Photo site: Zone 1, 14 Trees Vetale village

FAUNA OF 14 TREES



CLASSIFICATION OF TYPES OF FAUNA OBSERVED IN YEAR 2022 -23 AND 2023-24

FAUNAL DIVERSITY

Landscape of Vetale is dominated by scrub and grassland habitat on hill top. It has been observed that Raptors / Birds of prey species which are carnivorous in feeding habits such as Black Winged Kite, Harrier Sp, Oriental Honey Buzzard, Shikra, Common Kestrel occurring in the months of monsoon and winter. In the winter, the frequency of occurrence is high, this is possibly because scrub and grassland get dried, which makes them find and locate small birds, mammals. Also, the presence of small farm-fields attracts smaller birds, which also provide food to the raptors. For the last 3 years, the occurrence of raptors in the Vetale catchment has been stable.

Pugmarks of Leopard and Canid species were also observed at the site.

A) FAUNAL SPECIES DISCOVERED

Amongst the total of 127 species, 30 new fauna species discovered throughout April 2023 to March 2024 are as below:

	FAUNA SPECIES COMMON NAME AND SCIENTIFIC NAME	TYPE
1.	Ashy Drongo *Dicrurus leucophaeus*	Bird
2.	Blyth's Reed Warbler *Acrocephalus dumetorum*	Bird
3.	Brahminy Starling *Sturnia pagodarum*	Bird
4.	Bush Quail sp.	Bird
5.	Crested Bunting *Melophus lathami*	Bird
6.	Dusky Crag Martin *Ptyonoprogne concolor*	Bird
7.	Eurasian Roller *Coracias garrulus*	Bird
8.	Grey-breasted Prinia *Prinia hodgsonii*	Bird
9.	Harrier sp.	Bird
10.	Indian Nightjar *Caprimulgus asiaticus*	Bird
11.	Lark sp.	Bird
12.	Montague's Harrier *Circus pygargus*	Bird
13.	Oriental Bushlark *Mirafra erythroptera*	Bird
14.	Oriental Honey Buzzard *Pernis ptilorhynchus*	Bird
15.	Oriental Skylark *Alauda gulgula*	Bird
16.	Pipit sp.	Bird
17.	Yellow Browed Bulbul *Acritillas indica*	Bird
18.	Yellow Crowned Woodpecker *Leiopicus mahrattensis*	Bird
19.	African Babul Blue *Azanus jesous*	Butterfly
20.	Common Four Ring *Ypthima huebneri*	Butterfly
21.	Forget Me Not *Catochrysops strabo*	Butterfly
22.	Indian Cabbage White *Pieris canidia*	Butterfly
23.	Indian Cupid *Everes lacturnus*	Butterfly
24.	Red Peirrot *Talicada nyseus*	Butterfly
25.	Small Orange Tip *Anthocharis cardamines*	Butterfly
26.	Yellow Pansy *Junonia hierta*	Butterfly
27.	Yellow Banded Wasp	Wasp

Similar to the floral diversity, even the faunal diversity plays an integral role in building and enriching the ecological interrelationships. Presence of some faunal species also indicates certain ecological aspects like abundance of freshwater, clean air and nutrients availability. Total number of new species observed this year were: Birds-18, Butterflies-8, Wasp-1, Mammals (pugmarks observed, hence considered only an indirect evidence).

We have observed various fauna like mammals, birds, insects, reptiles, amphibians, chilopoda and arachnids (spiders) which reflect a complex food web which indicates a healthy ecosystem. Insects are key indicators of changes in soil, water, and air quality. The majority of insects, including beetles, ants, honey bees, and butterflies are sensitive to even the slightest environmental changes.

Some of the new faunal species observed this year (scrub and grassland)

Image: Pipit sp. (left) and Crested Bunting

B) CLASSIFICATION: TYPES OF FAUNA

TOTAL NUMBER OF SPECIES OF TYPES OF FAUNA AT 14 TREES

The above diagram reflects the total number of faunal species observed at 14 Trees in the last 4 years as a part of the biodiversity survey: September 2020 to March 2024.

113 SPECIES OF BIRDS OBSERVED AT 14 TREES (2020-2024)

A few of them are Indian Golden Oriole, Common Iora, Sunbird, Indian White-eye, Common Kingfisher

Butterflies and birds are nature's catalysts to biodiversity and food production through the process of pollination.

59 SPECIES OF BUTTERFLIES OBSERVED AT 14 TREES (2020-2024)

A few of them are Chocolate Pansy, Common Grass Yellow, Common Cerulean, Evening Brown

ZONE WISE OBSERVATIONS

10 years restoration: Zone 1

Zone 1 is a semi-restored zone with plantations of 10+ years old, where different microhabitats are seen in the form of stream, pond, cultivation area and scrub land that is now transformed into a Dry Deciduous Forest, which is now a primary habitat.

Habitat diversity and developed forest is attracting numerous bird & butterfly species. Species count is highest in Zone 1, with 72 species recorded during the survey conducted from April 2023 to March 2024. Zone 1 lies on hill slope and valley bottom area, while other 5 zones are mainly on hill tops and slopes, where habitats vary.

Scrub Hill slope: Zone 5

Ponds, hill slope: Zone 6

Except Zone 2, the rest of the 4 zones have a plantation of 1 or 2 years old. Effect of which, mixed diversity of bird species is observed, i.e. species belonging to scrub and grassland and species belonging to forest.

Presence of water bodies, farm-fields, cultivation, stream adds to the habitat diversity. Zone 6 shows a comparatively higher number of species, while Zone 3 & 5 show a similar number of species, because of the presence of more scrub land habitat species.

LANDSCAPE, HABITATS & BIO-INDICATORS

Insectivorous birds, which occur in scrub and grassland habitat indicates availability of smaller invertebrates, eggs and larvae of insects which indicates a stable food chain.

Occurrence of **Yellow Browed Bulbul,** endemic species to the Western Ghats, found in moist to semi- evergreen forests, indicates that site of 14 Trees: the base camp is transforming towards **Moist- Deciduous Forest.** This species was observed for the first time in the last 3 years!

The biomass and grass cover is observed more on the left side of the fence (protected area under restoration) whereas right is a barren land.

Snakes serve as touchstones of unpolluted, undisturbed habitats with availability of healthy plants and prey.

Species of birds viz, **Common Kingfisher**, which is observed near water-bodies, feeding primarily on fishes serve as good indicators of freshwater community health.

In the season of monsoon, amphibians, were observed in good numbers, such as **Indian Bull Frog, Indian Burrowing Frog, Asian Toad etc. They too are bioindicators of a healthy aquatic and terrestrial ecosystems.**

Pond and stream side vegetation areas are seen dominated with Pongamia pinnata (Karanj), Bamboo species, Saccharum spontaneum (Kamis),Persicaria glabra(Sheral) etc which are habitat specific plants and associated with other plant species like Azadirachta indica(Kadulimb),Vitex negundo(Nirgudi) etc.

Weedy growth of invasive plants species of Chromolaena odorata(Raanmari),Lantana camara(Tantani),Alternanthera ficoidea(Chandani gavat/Chibuk kata),Parthenium hysterophorus(Gajar gavat) are also having its occurrence at places occasionally. It's important to manage invasive plants and this challenge is being tackled with the help of the experts and volunteers at 14 Trees.

SEEN-UNSEEN DISCOVERIES AT 14 TREES

SNAKE MOULD NEAR A POND

PUGMARK OF A LEOPARD

SNAKE MOULD

PUGMARK OF CANID Species (Wild)

PUPA OF AN INSECT

AMAZING FACTS OF FLORA & FAUNA

Nature's very own recyclers!

Rock-growing lichens play an important role in nature's plan by converting rocks into soil. They produce a weak acid which slowly dissolves the minerals, forming tiny cracks in the surface of the rock.

Lichens observed on the rocks also in the open-scrub and grasslands of 14 Trees.

Architects of Nature!

Harvester Ants nests have two or more walls that keep rain water from entering the nest and maintain the indoor temperature, as well. At times, such walls are raised from the ground, forming a circular structure; these kinds of nests can be largely seen in areas with heavy rainfall.

Nature's engineers- Baya Weaver's knots!

The nests are built by the male and they are the only birds recorded with the ability to tie knots to make complex nest structures!

The female selects the nest she likes the most.Nests are mainly woven on the tree branches hanging above the water body in order to protect themselves from predators.

FOOD FOREST & NURSERY

14 Trees Foundation

LOGO

14 TREES NURSERY & FOOD FOREST

LOCAL (NATIVE) SPECIES VARIETY

The purpose of the in-house 14 Trees Nursery is to create a local and self-sufficient supply of saplings and food in the food forest which helps save transport emissions, generates local employment and healthy food without the use of chemicals like pesticides. Hence, promote sustainable food practices.

‘8 NATURALS’ NURSERY OF 14 TREES MAINTAINS AROUND 450 TYPES OF PLANT SPECIES

VARIETY IN FRUIT TREE SPECIES

No. of Varieties

- Lemon - 22
- Mango - 20
- Guava - 7
- Apple - 5
- Jackfruit - 3
- Tamrind - 3

Other fruits:

- Orange
- Bor
- Avla
- Jambhul
- Kavath
- Banana

HARVESTED MANGOES AT 14 TREES

Red Banana

‘Mahalunga’ Lemon

14 Trees
Foundation

"Nature is not a place to visit, it is Home."
-Gary Synder

14 Trees
Foundation

In frame: 'Common Grass Yellow' butterflies mating

End of Report